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2025-yil 4-variant

1. 5 ga bo'lganda 4 qoldiq qoladigan 1000 gacha bo'lgan barcha natural sonlar yig'indisini toping.
A) 154660
B) 1972310
C) 111111
D) 100300

2. $2019 - 2x = 2021$ tenglamani yeching.
A) 1
B) 4
C) 11
D) -1

3. Soddalashtiring: $(10 - 4a) : a - 2$
A) 1
B) $\frac{10}{a} - 6$
C) $a - 1$
D) $\frac{2a}{3} - 1$

4. $x^4 + 12x^2 - 64 = 0$ tenglama nechta haqiqiy ildizga ega?
A) 1 ta
B) 2 ta
C) 4 ta
D) 3 ta

5. $x^4 - 5x^2 - 24 = 0$ tenglamaning ildizlari ko'paytmasini toping.
A) 8
B) -8
C) 24
D) -24

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6. Hisoblang:

$$(-2{,}4):(-0{,}8)+7$$

A) 10
B) 40
C) 4
D) 16

7. $\begin{cases} x^2+3=2xy \\ 6x^2-11y^2=10 \end{cases}$ tenglamar sistemasida y ning qabul qilishi mumkin bo'lgan barcha yechimlari ko'paytmasini toping.

A) -9
B) 36
C) -4
D) 4

8. Tenglamaning eng katta manfiy va eng kichik musbat yechimlari yig'indisini toping.

$$(\cos 2x-1)\cdot(2\sin x-\sqrt{3})=0$$

A) $\frac{\pi}{3}$
B) $-\frac{\pi}{3}$
C) $-\frac{2\pi}{3}$
D) $\frac{2\pi}{3}$

9. $(2\sin 2x-\sqrt{3})(2\cos 2x-1)=0$ tenglamaning $[-\pi;2\pi]$ oraliqdagi yechimlari nechta.

A) 11
B) 9
C) 10
D) 12

10. $f(x)=\sqrt{\frac{x-1}{x+1}}$ funksiyaning $x_0=2$ nuqtadagi hosilasini toping.

A) $\frac{\sqrt{3}}{9}$
B) $-\frac{\sqrt{3}}{9}$
C) $-\frac{1}{3}$
D) $\frac{\sqrt{3}+1}{3}$

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11. $100 \cdot 4^{12} \cdot 5^{23}$ nechta nol bilan tugaydi?

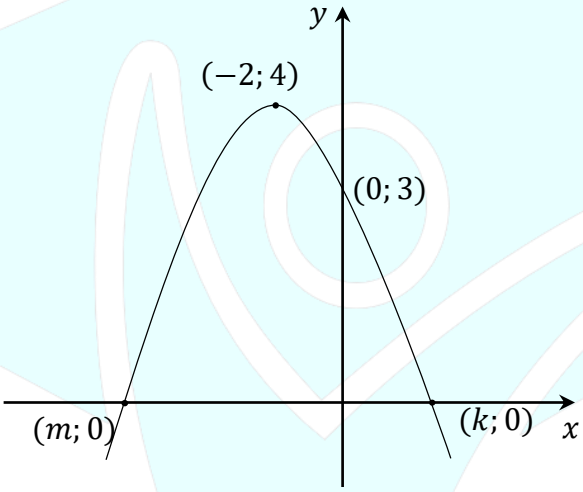
A)25

B) 23

C) 30

D) 26

12. Kvadrat funksiya Ox o'qini $(m; 0)$ va $(k; 0)$ nuqtalarda kesib o'tadi va parabola uchining koordinatalari $(-2; 4)$ bo'lsa, $m + k$ ni toping.



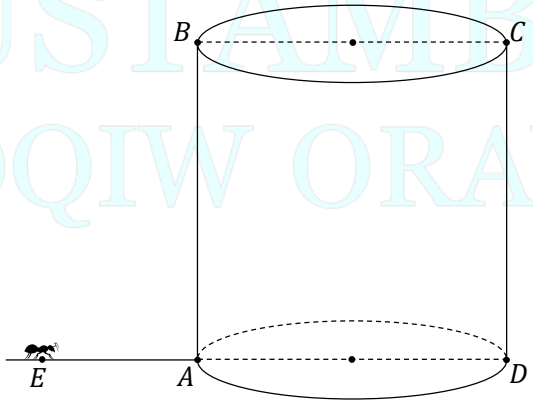
A) -4

B) -6

C) 1

D) 6

13. Rasmda E nuqtada shumoli ko'rsatilgan va uning yonida silindrsimon idish turibdi. Shumolidan idishgacha eng qisqa masofa 8 cm ga teng. Silindr idishning o'q kesimi $ABCD$ kvadratdan iborat. Silindr idishning asosining diametri 16 cm teng. Shumoli E nuqtadan D nuqtagacha harakatlangandagi eng qisqa masofani toping? ($\pi \approx 3$ *deb olinsin*).



A) $24 + 4\sqrt{3}$

B) $6 + 8\sqrt{3}$

C) 32
D) $8\sqrt{3}+16$

14. Muntazam to'rtburchakli kesik piramidaning kichik asosi yuzi 16 ga, katta asosi yuzi 81 ga va piramidaga shar ichki chizilgan. Shar hajmini toping.
A) 576π
B) 144π
C) 288π
D) 36π

15. Poyezdga 18 kishi chiqdi so'ng 6 kishi tushib qoldi va so'ng poyezda 31 ta odam qoldi. Dastlab poyezda nechta odam bo'lgan?
A) 38
B) 19
C) 17
D) 15

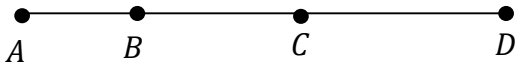
16. $\sqrt[8]{3\sqrt{3\sqrt{3^a}}}$ ifoda natural son bo'ladigan a ning eng kichik natural qiymatini toping.
A) 8
B) 12
C) 24
D) 26

17. Tenglamani yeching: $144:x=36$
A) 4
B) 10
C) 8
D) 16

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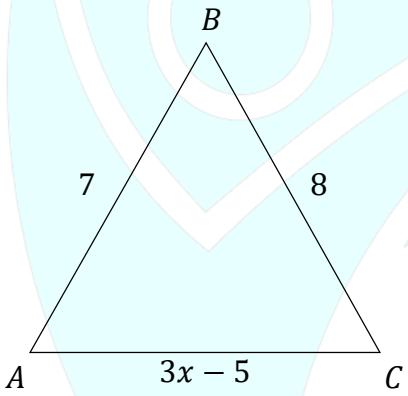
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18. A, B, C, D nuqtalar berilgan AC kesmaning uzunligi 25 cm, BD kesma uzunligi 27 cm, AD kesma uzunligi $45\,cm$ ga teng bo'lsa, BC kesma uzunligini toping.



A) 11
B) 8
C) 10
D) 7

19. ABC uchburchakning perimetri 22 ga teng bo'lsa, x ning qiymatini toping(Rasmga qarang).

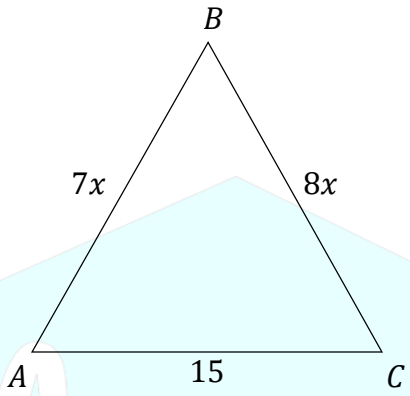


A) 1
B) 4
C) 3
D) 5

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20. ABC uchburchakning perimetri 45 ga teng bo'lsa, x ning qiymatini toping. (Rasmga qarang).



A) 1
B) 2
C) 3
D) 4

21. Quyidagi jadvalda mahsulotlarning narxlari berilgan. Jami mahsulotning narxini(so'm) toping.

Nomi	1 kg narxi (so'm)	Harid qilinadigan kg
Pomidor	9000	3 kg
Baklajon	13000	4 kg
Bodring	8000	5 kg

A) 120000
B) 119000
C) 103000
D) 140000

22. 3 ta qopning jami og'irligi 84 kilogramm. Agar birinchi qop ikkinchi qopdan 5 kilogrammga og'ir, ikkinchi qop esa uchinchi qopdan 4 kilogrammga yengil bo'lsa, uchinchi qop og'irligini toping.
A) 11
B) 25
C) 7
D) 29

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23. Hisoblang:

$$(-1{,}2):(-0{,}2)+5$$

A) 10
B) 11
C) 12
D) 13

24. Kubning asosining yuzi 64 ga teng bo'lsa, kubning to'la sirtining yuzini toping.
A) 100
B) 400
C) 384
D) 272

25. Sinfda 24 ta o'quvchi bor. Sinf o'quvchilar orasidan sinf sardorini tanlashning nechta usuli bor?
A) 12
B) 40
C) 24
D) 72

26. $10\ metr/sekund$ necha $km/sekund$ ga teng.
A) 0,1
B) 0,01
C) 0,001
D) 0,0001

27. Agar $x=-5{,}25$ va $y=-2{,}25$ bo'lsa, x^2-xy+y^2-xy ni hisoblang
A) 4
B) 16
C) 12
D) 9

28. Trapetsiyaning kichik asosi 8 ga, katta asosi 10 ga teng. Asoslariga parallel va yuzini teng ikkiga bo'luvchi kesmaning uzunligini toping.
A) $\sqrt{82}$
B) $\sqrt{21}$
C) 9
D) $10\sqrt{2}$

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29. $f(x) = 2x^2 + 3$ funksiyaning grafigi qaysi choraklardan o'tadi.

A) I, III

B) II, III

C) I, II

D) I, IV

30. $\frac{52}{17} + \frac{39}{19} - \frac{85}{21}$ ifodaning qiymati qaysi oraliqda bo'ladi.

A) $(0; 1)$

B) $(2; 3)$

C) $(1; 2)$

D) $(3; 4)$

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